One Machine: the verified build

OES end-to-end model, 28 October 2026. This is a selected mathematical build, replayed from matrices. It is not a source admission and not a change to the canon matrices.

AUTH = NOT_PASS · ScientificGold = FALSE · TIME_BIND = UNBOUND · DYN_BIND = NOT_INSTALLED · SCR6 = NOT_RUN_OPERAND_ABSENT · PredictionFreeze = NOT_TRIGGERED. No gate moves.

Custody.

- Repository `sachinkaria24-byte/OES`, branch `claude/lagrangian-spectral-spacetime-model-t2253b`.
- `python verify.py` replays all 18 layers: **320 checks**. Of these, 300 are statements that hold: THEOREM 214, ARITHMETIC 26, OBSTRUCTION 22, CONDITIONAL 15, UNFORMED 15, SELECTED_REFERENCE 6, REMOVED 2. The other 20 are REPORTED: they are recorded, not counted.
- `python -m pytest` runs 22 tests.

0. What this build adds

The 14:16 snapshot (`one-machine-20261028.pdf`, `oes-model-20261028.zip`) checks its joins by comparing typed constants (`251 === 251`). Here every statement is computed from explicit matrices by a check that can fail. On top of everything the snapshot covers, this build proves:

1. **The scalar walk recurrence, formed.** $W_0 \ldots W_{90}$ are exact from two independent engines, and $W_0 \ldots W_{600}$ are known modulo primes. The annihilator is one shift operator of **order 16 and degree 58**, the same mod four primes. Built from $W_0 \ldots W_{500}$, it predicts $W_{501} \ldots W_{600}$. Its characteristic polynomial factors exactly over $\mathbb{Q}$.
2. **The band and the walks agree.** The 11 critical families of the E8 band (243,600 points, Morse sum 0) have exactly the 11 in-band characteristic roots of that operator as their values.
3. **Density on the nine-torus.** The nine positive bath energies have distinct square-free parts, so their frequencies are linearly independent over $\mathbb{Q}$ and their flow is dense on T^9. The snapshot left this open.
4. **All eleven Hessians.** Recomputing them reproduces every coherent coefficient the snapshot prints. The snapshot said they were not recomputed.
5. **The walk asymptotics.** $W_n = (16/\pi^4)\cdot 240^n \cdot n^{-4} \cdot (1 - 2/n + (23/15)/n^2 + O(n^{-3}))$, derived from the 7-design moments alone and followed by the exact $W_{50} \ldots W_{90}$.
6. **Sound, light, frequency and phase on the K5 edges.** The Hodge Laplacian is $5 \cdot I$, so sound and light share one frequency. The canon's Hodge block A is the curl, and the two light helicities are the icosahedral 3 and 3′. Agni = $5I - G$ is the light operator Curl^2.
7. **The literal five-dimensional operand.** The printed chart T_6 preserves the contrast lattice V_c. $T_6|V_c$ is a 5×5 integer operator with determinant 31, which explains $155 = 5 \times 31$.
8. **Corrections** to the snapshot, the Locked Memory handover and Court U19.81 (§6).

1. The carrier and its Lagrangian (layers 1–3)

- The carrier has 15 atoms, the 5 vertices and 10 edges of K5. M is the signless incidence, and $D = [[0, M], [M^T, 0]]$.

- **spec $D^2 = \{0^5, 3^8, 8^2\}$**, with Tr $D^2 = 40$ and Tr $D^4 = 200$. The kernel ker D = P5 is the 5-dimensional irreducible [3,2] of S5.
- The Lagrangian is $S = \int \tfrac{1}{2}|\dot{\psi}|^2 - \tfrac{1}{2}\langle\psi, D^2\psi\rangle$, with no other term. The frequencies are 0, √3 and 2√2. Their squared ratio 8/3 is not a rational square.
- The signed boundary B (the canon's `boundary_oriented`) satisfies $MM^T + BB^T = 8I$: the carrier's vertex frequencies are the complements of the sound frequencies (§4).
- The E8 bath is $\Delta = 240I - A$ on the root lattice. Its band is [0, 256], its long-wave dispersion is $\varepsilon(k) = 30|k|^2 - (3/2)|k|^4 + (1/24)|k|^6 + O(|k|^8)$ (exact, from the 7-design), and its sound speed is √30.

2. The native action and the operands (layers 4, 10–12)

- **Source.** q = (a, b) ∈ P5 ⊕ P5, with M_s = diag(W/4, W/4). At the origin the source obeys **$\ddot{q} + 4q = 0$**, so $\omega_0 = 2$.
- **Port.** g = W_AS(a ∧ b), a quadratic map onto the 6-dimensional cycle space L6. The sharp quartic constant is $Q_4 \le (40/9)\rho^2$, and it is attained.
- **Bath.** The six-site clique has A_bb = 235P_u + 241P_c and BB* = 183P_c + 495P_u.
- **Literal operands.** The native file (SHA-256 73e07429…) was recovered, and every canon identity on it replays. The literal port, in edge coordinates (`Fedge_native_inverse · WAS_native`), is S5-equivariant.
- **The printed 6×6 chart T_6** (det 155):
  - Every column sums to 5, so T_6 preserves $V_c = \{x \in \mathbb{Z}^6 : \Sigma x = 0\}$.
  - On V_c it is the integer matrix [[1,−1,1,0,−1], [1,−1,3,−2,−2], [1,−2,4,−3,0], [1,0,3,−1,0], [2,0,1,−1,0]], with det 31 and irreducible characteristic polynomial $x^5 - 3x^4 + 12x^3 - 20x^2 + 31x - 31$. This is the candidate for the missing five-dimensional operand; the owner must confirm it.
  - The largest |cofactor| is 68. $4805 = \text{cofactor}_{44}(TT^T) = 21^2 + 48^2 + 43^2 + 11^2 + 3^2 + 9^2$, a sum of squared cofactors and not a minor.
  - The handover's "literal" T5×5 = $(2I_4 - C_{A4}) \oplus [5]$ is a construction: it is not in the file and not a block of T_6.
- **Glue.** E8 ⊃ A4 ⊕ A4′ has index 5, with glue $x \mapsto \pm 2x$. The carrier's atoms are the A4 shadows of the 200 mixed roots, and 248 = (24,1) + (1,24) + four bifundamentals.
- **Binding constraints.**
  - The only atom-diagonal symmetry of D is the global phase. ker D carries an exact U(5).
  - The port breaks the dark O(5) completely. The minimal anomaly-free completion is $5 + \Lambda^2 5 = 15$ states, which is one generation per carrier.
  - The generation multiplicity is an operand of DYN_BIND.

3. The walk recurrence and the band geometry (layers 14, 16, 18)

Data.

- The Weyl-orbit DP gives exact integers $W_0 \ldots W_{90}$. All 56 canon values $W_0 \ldots W_{55}$ are reproduced.
- A second engine uses E8 = D8 ∪ (D8 + s) and $\lambda = 4e_2(c) + 128\prod\cos(\theta_i/2) + 128\prod\sin(\theta_i/2)$, with Hermite–Gauss quadrature over F_p. It gives $W_0 \ldots W_{600}$ mod two primes and $W_0 \ldots W_{500}$ mod two more.
- The two engines agree on all 91 exact terms.

Exclusion and the operator.

- Every recurrence with $(r+1)(d+2) \le 455$ is excluded. In particular, order 11 needs degree ≥ 49.
- The first kernels appear at shape (23, 17). That kernel is 2-dimensional mod both 600-term primes, with 146 surplus equations.
- The greatest common right divisor of the kernel operators is one operator **L_{16}, of order 16 and degree 58**, the same mod four primes.

Characteristic polynomial (exact over $\mathbb{Q}$).

> $(z-240)(z-16)(z+3)(z+8)(z+10)(z+11)(z+185/16)(z+320/27)(z+12)^2(z+16) \cdot (z-340)(z+660)(z+20)(z+35)(z+120/7)$

The local exponent is $-4 = -\dim/2$ at every simple root, and the indicial polynomial at -12 is $(\alpha+4)(\alpha+5)$. The first factor group contains the 11 roots in the band $[-16, 240]$; it is the sibling's "order 11 with $(x+12)^2$". The five roots outside the band are why the operator has order 16.

The critical census (vertices and edges of the alcove). The table lists $\lambda = 240 - \varepsilon$, the number of points, the Morse index of ε, and the exact det of the Hessian of ε:

family	λ	points	index	det H
E8	240	1	0	60^8
E7+A1	16	120	1	$-2^{16}\cdot 3^7\cdot 13$
E6+A2	−3	2,240	2	$2^6\cdot 3^8\cdot 7^2$
D5+A3	−8	15,120	3	$-2^{16}\cdot 3^3$
A4+A4	−10	48,384	4	5^8
A5+A2+A1	−11	80,640	5	$-2^6\cdot 3^7$
A6+A1 (edge)	−185/16	69,120	6	$3^7\cdot 5^{15}/2^{30}$
A7 (edge)	−320/27	17,280	7	$-2^{16}\cdot 5^{15}/3^{23}$
A7+A1	−12	8,640	7	-2^{16}
A8	−12	1,920	8	3^8
D8	−16	135	8	2^{16}

- In total there are 243,600 points, and the alternating sum $\Sigma(-1)^{\wedge}\text{index}\cdot\text{points} = 0 = \chi(T^8)$.
- The coherent coefficients points · i^index / √|det H| reproduce the snapshot's printed values. The 11 values of λ are exactly the in-band characteristic roots above: band geometry and walk data are one statement.
- The band has no critical value in (0, 224).

Asymptotics. At the maximum, $\log(1 - \varepsilon/240) = -s/8 - s^2/640 - s^3/23040$ with $s = |k|^2$. Laplace's method then gives

> $W_n = (16/\pi^4)\cdot 240^n \cdot n^{-4}\cdot(1 - 2/n + (23/15)/n^2 + O(n^{-3}))$.

The exact $W_{50}\ldots W_{90}$ follow it. The residual $((R_n - 1 + 2/n)n^2 - 23/15)$ falls from -0.0069 at $n = 50$ to -0.0038 at $n = 90$.

Density on T^9.

- The nine positive critical energies are 224, 243, 248, 250, 251, 4025/16, 6800/27, 252 and 256 (the two families at 252 share one).

- Their frequencies √ε have the distinct square-free parts 14, 3, 62, 10, 251, 161, 51, 7, 1.
- By Besicovitch these square roots are linearly independent over ℚ. By Kronecker, the joint flow is dense on the nine-torus.

4. Sound, light, frequency and phase on the K5 edges (layer 17)

The edges carry one exact Hodge complex: the gradient $d_0 = B^T$, and the curl Curl = $\star d_1$ through the ten triangles and the Hodge star of the 4-simplex.

- **Sound** is im d_0, the 4 longitudinal modes. Its operator $BB^T = 5I - J = \frac{1}{2}\Sigma_{A4}\ \alpha\alpha^T$ is the long-wave Hessian of the carrier's own root lattice.
- **Light** is ker d_0^T, the 6 transverse modes. This is the cycle space L6, the port space. Curl is symmetric with eigenvalues ±√5 on light. In the canon's fundamental-cycle basis it is **exactly −A**, the Hodge block of layer 1.
- **Frequency.** $d_0 d_0^T + \text{Curl}^2 = 5 \cdot I_{10}$, so sound and light are isochronous at √5.
- **Agni.** Agni = 5I − G = Curl² is the light operator, and G = B^TB is the sound operator.
- **Phase.**
  - Sound carries the C5 phases ±72° and ±144°, and none at 0.
  - The two helicities are swapped by odd permutations. Under A5 they are the icosahedral 3 and 3′, with 5-cycle characters 1 − φ and φ; sound is the A5 4.
  - No S5- or A5-symmetric medium turns sound into light. The C5 clock allows exactly 4 phase-matched parameters, and the phase-0 light modes stay dark.
  - With a C5 coupling, conversion is resonant and complete (CONDITIONAL).
- **The chain.**
  - The dark source at ω = 2 drives light through the quadratic port at 0 and 4, doubling frequency and phase. The port's kernel is the A5 4.
  - The adapter carries light to the boundary. There the bath takes it as sound in the smooth acoustic window, at k ≈ 0.740, with phase and group velocities 0.986 and 0.959 of √30.
  - On the carrier's plane the bath speed² splits as 30 = 5 (A4) + 25 (the mixed roots).
- **Frequency ladder.** 0 < √3 < 2 < √5 < 2√2 < 4 < √224 < 16.
  - The squares 3, 4, 5, 8, 16 divide 4320, and no bath energy does.
  - $4b^2 = 5a^2$ has no solution, so the source and sound frequencies are incommensurable.

Making "sound" and "light" physical needs DYN_BIND and TIME_BIND. The invariant content is the set of ratios.

5. The 14:16 snapshot, rebuilt (layer 18)

- **Agni–clock–doubling.** Agni = Curl², as in §4. The edge doubling T (10×20) has $TT^T = 2I$, is isotropic for the pair signature, and has rank BT = 4.
- **Five-port site block** (SELECTED_REFERENCE): masses 1 and 3, springs [[2,−1],[−1,2]].
  - Its moments are 0, 0, 0, 1/3, 0, −8/9, 0, 55/27, with recurrence $H_{k+4} + (8/3)H_{k+2} + H_k = 0$.
  - Its controllability determinant is −1/3 and its observability determinant −1/27.
  - Its Hankel determinant is 1/81 (6561 cleared), and its frequencies² are (4 ± √7)/3.
- **Cube.**
  - The eight labels 580…079 form the 3-cube, with Laplacian spectrum 0, 2^3, 4^3, 6.

- The middle operator $L(6I - L)$ satisfies $M^2 = 8M$ and has rank 6.
- The reader pair sees ranks 2 then 4; four third readers reach 6.
- The decoder has det -1, with noise gains 46 and 66.

- **Spanda** on the 31 subsets: 7 fixed, 24 moved, traces 31, 7, 31, 7, and trace-Hankel determinant 912. On the K5 edges the same double transposition keeps both helicities, with traces 0 on sound and -1 on each helicity.
- **4320-circle.** The day (order 24) and the pada (order 108) meet in the order-12 step, and the order-5 rotor misses the day.

6. Corrections to circulating documents

- **14:16 snapshot.**
  - $(I + S)/2$ is a Markov projector of rank 19 (7 fixed subsets plus 12 symmetrized pairs), not "onto those 7 forms".
  - The front table's "the recurrence order is 11" should read: order 16, with an in-band part of degree 11.
  - Its "not replayed" items (density on T^9, the Hessians) are settled above.
- **Locked Memory handover.**
  - The "15-edge K6 carrier" is the 15 atoms of K5.
  - An order-11, degree-5 exact recurrence is excluded. P25 annihilates only the quadrature sequence.
  - $(15/13)r^2$ should be $(15/16)r^2$.
  - "ker O ⊆ ker C" is reversed.
  - "H90 identified" is unsupported.
  - "Multi-frequency criterion verified" is true only as rebuilt.
  - The "literal" T5×5 is a construction.
  - The paper digest b5590733… matches none of the three One Machine PDFs on Drive (77fc5845…, ce509d99…, 714db33d…).
- **Court U19.81.**
  - The walk generator is wrong from W_4 on.
  - Its moment sequences are not positive definite.
  - Its uniform-sector moment formula is wrong.
  - Its Loewner certificate fails on the true moments.
  - Its 2-parameter clock reduction is impossible for the V_2 it names.

7. Gates

Gate	State	What closes it
AUTH / SCR6	NOT_PASS / operand absent	the owner's signature on the packaged antichain; the six SCR6 operands of a chosen physical child
TIME_BIND	UNBOUND	one authenticated clock datum (frequency ratios are calibration-invariant)
DYN_BIND	NOT_INSTALLED	the physical child, the generation multiplicity, the chirality index
PredictionFreeze	NOT_TRIGGERED	the owner freezes the register before new comparisons

Gate	State	What closes it
W_L	NOT_PASS by evidence	the recorded PMNS candidate is excluded ($\chi^2 = 10.6$ on 2 dof, $p = 0.005$)
T3081	UNOPENED	the maintainer side is replayed; the official host must be allowed
ScientificGold	FALSE	a frozen prediction, independent data and a preregistered test

8. What stays open

- The convexity collar inside (223.58, 224), which needs an analytic saddle cap plus a non-grid bound.
- The rational coefficients of L_{16}, and a proof that it annihilates W_n for every n.
- A preparation protocol matching the reported ε^(−1/2) and ε^(−1/3) bounds.
- The 8-dimensional dispersive constant.
- The six SCR6 operands, the 1401 → 1370 seams, the 502- and 237-dimensional operators, MPKG A_M, and the historical Agni, Soma, Suraj and Sawan matrices.
- The owner's choices: signature, physical child, clock datum, freeze.

Appendix: the eighteen layers

#	layer	checks	content
1	carrier	35	K5, D, spec, Lagrangian, exact motion, Petersen, Hodge block, edge tick
2	spectral	9	heat trace, ζ_D, spectral dimension, moments 15/40/200
3	spacetime	34	RP^2 pair, clock 125 → 120, E8 roots, walks, symbol, Green, clique coupling
4	native	19	source–port–bath action, W_AS, W_p, 40/9, radiation channel, nonlinear flow
5	observer	7	Kalman/Gramian readouts, Witten index −5
6	gravity	8	LLY curvature; spectral-action route (CONDITIONAL)
7	cosmology	7	FLRW with the carrier field (CONDITIONAL)
8	frontier	23	band-edge orbit 135, the 224 saddle, saddle law 15/16 d^2, collar survey, coil
9	joins	24	the maps that make the layers one object
10	literal	20	the recovered operand file, every identity replayed, the flow on the owner's operands
11	glue	11	E8 = D8 + s; A4 ⊕ A4′ with x ↦ ±2x
12	binding	12	DYN_BIND constraints
13	gates	12	TIME_BIND courts, register, W_L, T3081, AUTH package
14	recurrence	16	W_{90} exact, W_{600} mod p, exclusion, the order-16 operator, characteristic polynomial
15	court	19	Court U19.81 audited
16	handover	30	the Locked Memory handover audited; critical census; T_6
17	waves	18	sound, light, frequency and phase on the K5 edges
18	snapshot	16	the 14:16 snapshot rebuilt; T^9 density; Hessians; asymptotics

To replay: `pip install -r requirements.txt`, then `python verify.py` (about 2.5 minutes), then `python -m pytest -q`.